

Softwaregestützte Modellbildung und Simulation: Exercise 2

Principal Component Analysis (PCA)

Peleg Bar Sapir

June 4, 2026

Principal Component Analysis (PCA) is an application of Singular Value Decomposition (SVD) ...

The data set is ...

1. Load the data into a single NumPy array, using `numpy.loadtxt()`. Mathematically, we will call this data array **X**. Note that the **rows** of **X** are the separate *samples* (measurements), while its **columns** are the variables (features): *sepal length*, *sepal width*, *petal length* and *petal width*.
2. Center and normalize the data by subtracting from each **column** of **x** its mean, then dividing it by its standard deviation. Use the functions `numpy.mean()` and `numpy.std()`, respectively. Which axis should you use for these operations? Call the resulting matrix **B**.
3. Apply the SVD method to the matrix **B**:

```
1 U, S, VT = numpy.linalg.SVD(B, full_matrices=False)
```

(where **U**, **S**, **VT** are, respectively, the matrices **U**, **Σ**, and **V[†]** in the SVD of **B**)

4. In the terms of PCA, the columns of **U** are called the *Principal Components* (PC) of **B**. The number of principal components *N* is the number of variables in the data:

$$PC_1, PC_2, \dots, PC_N.$$

The corresponding Singular Values are the diagonal elements of the matrix **Σ**, denoted as

$$\sigma_1, \sigma_2, \dots, \sigma_N.$$

The singular values are ordered by absolute value in descending order. The variance in PC_i is given by

$$\text{Var}_i = \frac{\sigma_i^2}{r-1},$$

where r is the number of *samples* and $i = 1, 2, \dots, N$.

Plot the variance for each PC in descending order.

5. What do we get if we divide each Var_i by the sum of all variance terms? Plot these values as function of i . (**hint**: what is the sum of these terms?)
6. Plot the *cumulative sum* of the values calculated in the previous step, again against i (use the function `numpy.cumsum()`). Does the cumulative sum seem to converge, and to which limit? Explain briefly.
7. ...